Routing with OpenStreetMap Data

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National Large-scale Research Center of the Helmholtz Association

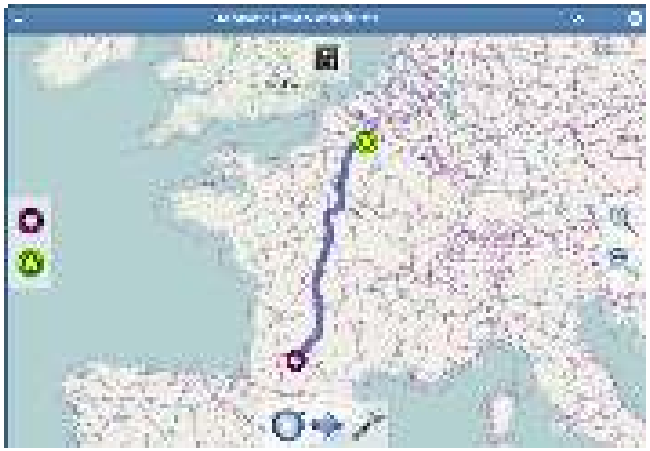
-Part I-
MoNav



What's MoNav?

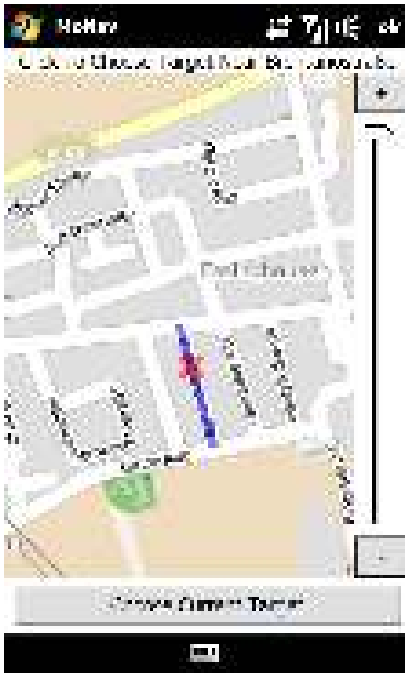
- Routing software for
  - Mobile Devices
  - Desktop PCs
- Fast Route planning
  - $<$200ms on mobile devices
- Route planning service





MoNav 0.1

- Fast route planning
- Address search
- Tile renderer
  - 4 GB for Germany
- Rusty UI



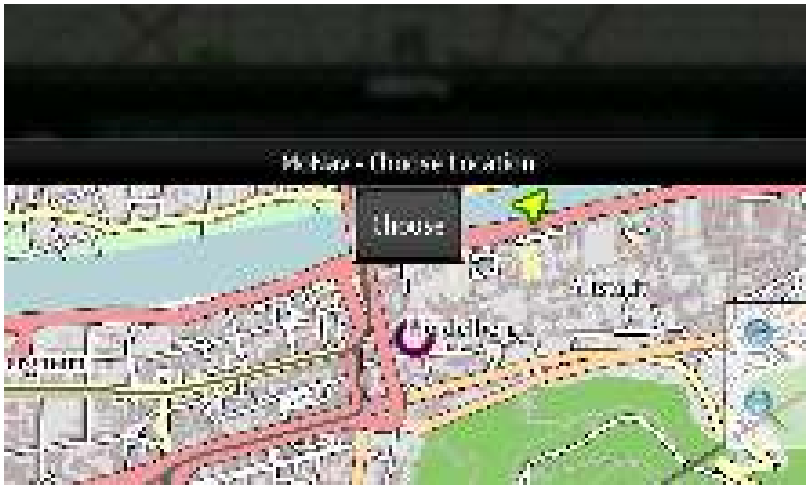


GUI



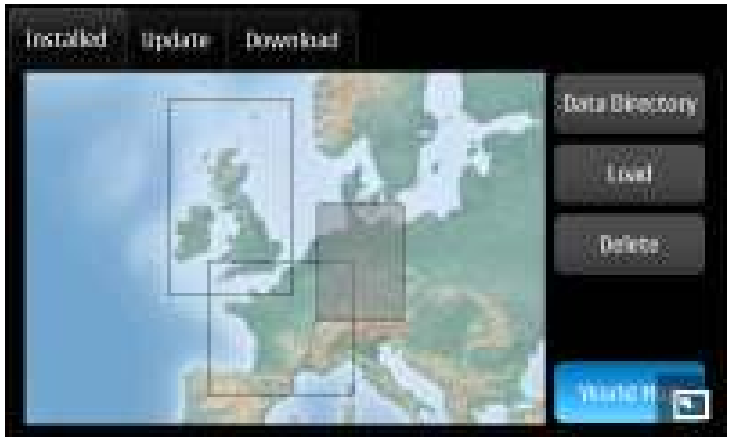

Map Packages
Map Modules
General
GPS Lookup

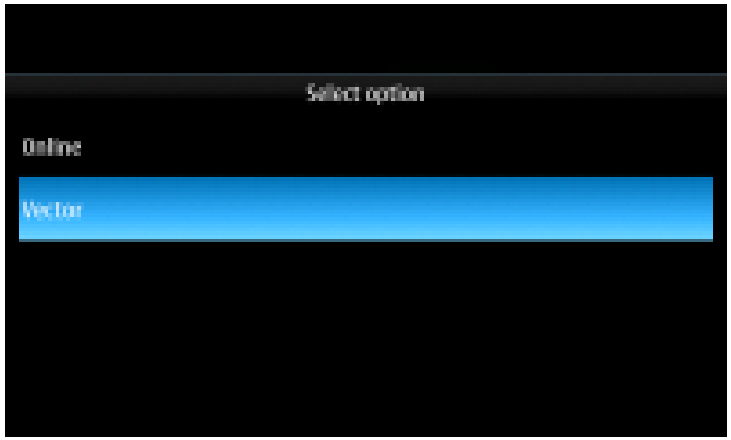

Installed
Update
Download
Data Directory
Load
Delete


Choose

Map Management

- Support for multiple map packages
- Easy selection of individual maps
- Automatic map download
- Automatic updates





Offline Vector Rendering

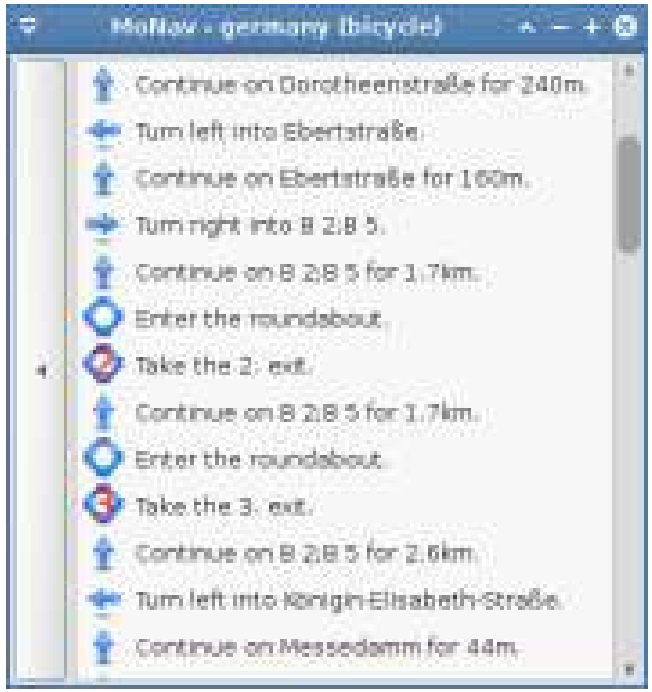
- Small amount of data
- DPI independent
- Currently two modules in progress:
  - libosmscout Renderer
  - Qtile Renderer



Turn Instructions

- Text based instructions
- No: "Do a U-Turn!"







PBF Support

- Binary OpenStreetMap Format
- Based on Googles Protocol Buffers
- Reading 10 times faster
- Smaller files
- Same information as OpenStreetMap XML files
- MoNav: Early support for PBF (September 2010)

MoNav as background service

- MoNav is GPL
- Distinct Routing Service
  - also GPL
  - But: Interface is LGPL
- Used by:
  - KDE's MARBLE





Bilder von: http://nienhueser.de/blog/

MoNav

- ... and many more improvements
- User:
  - 0.3 Released
  - Download and use
- Developers:
  - Routing service as LGPL
  - Still looking for developers and testers



-Part II-
Open Source Routing Machine



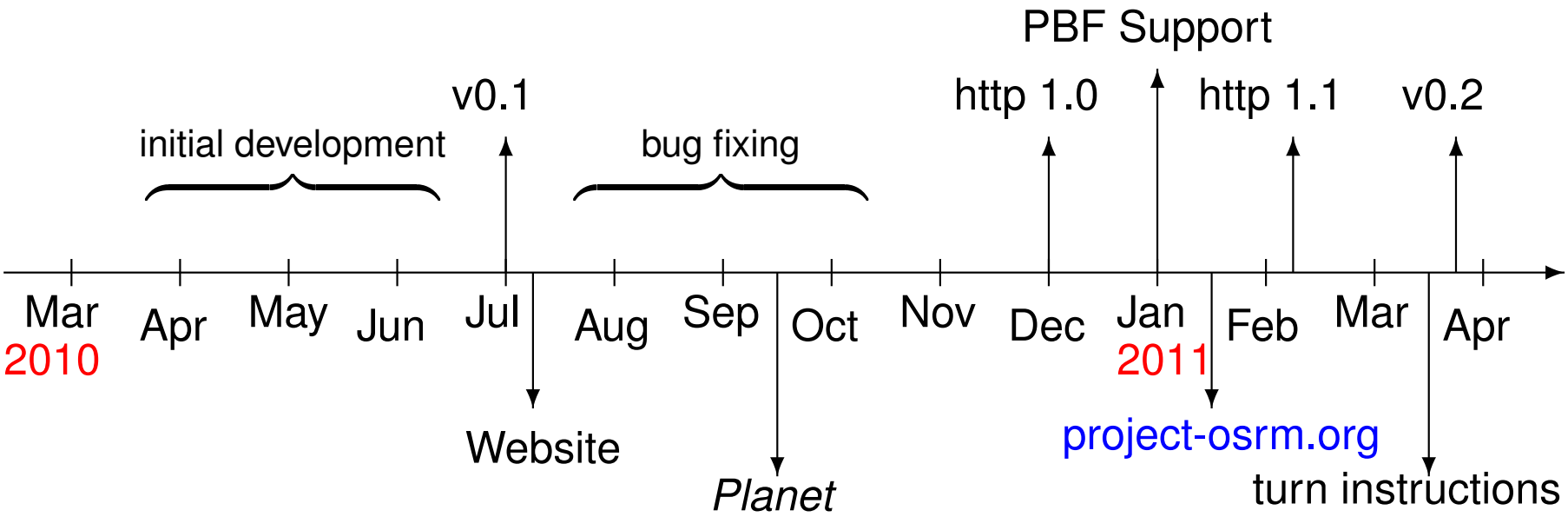
What's OSRM?

Server-based Routing Engine

- Common algorithmic base with Monav
- Fast (<100ms incl. network)
- AJAX web GUI
- approx. 7.000 lines of C++



Die first 12 months



The past 6 months

- Ajax GUI by Pascal Neis
- Draggable Routes
  Beta: draggable via points
- Permanent Links
- GPX/JSON output (KML deprecated)
- Compression of tranferred data
- Route generalization

Online Demo

- `http://map.project-osrm.org`
  hosted by Geofabrik.de







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Lots of public attention: 250.000 routes in first two weeks

Demo

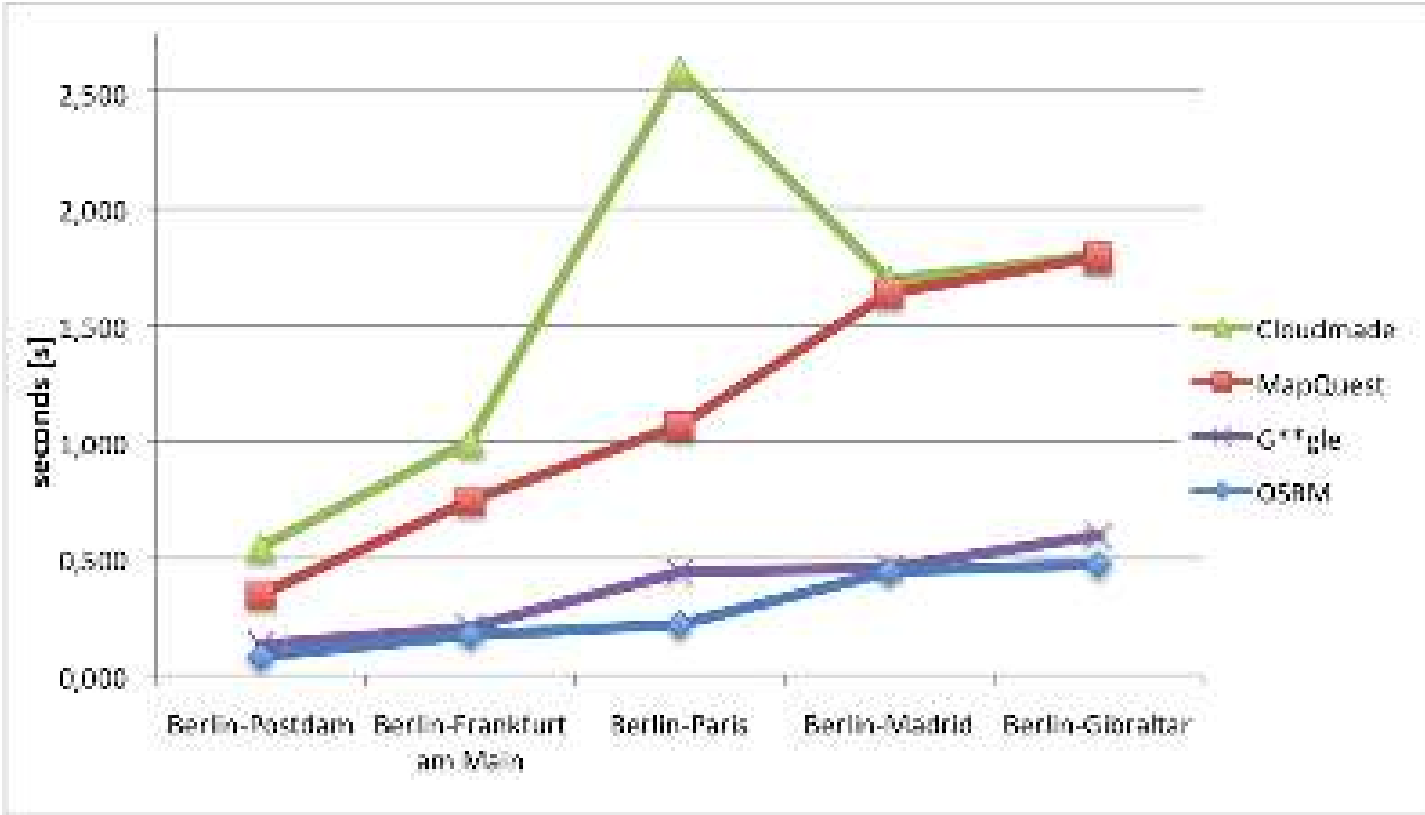
Demo online:

http://routingdemo.geofabrik.de



What does the public say

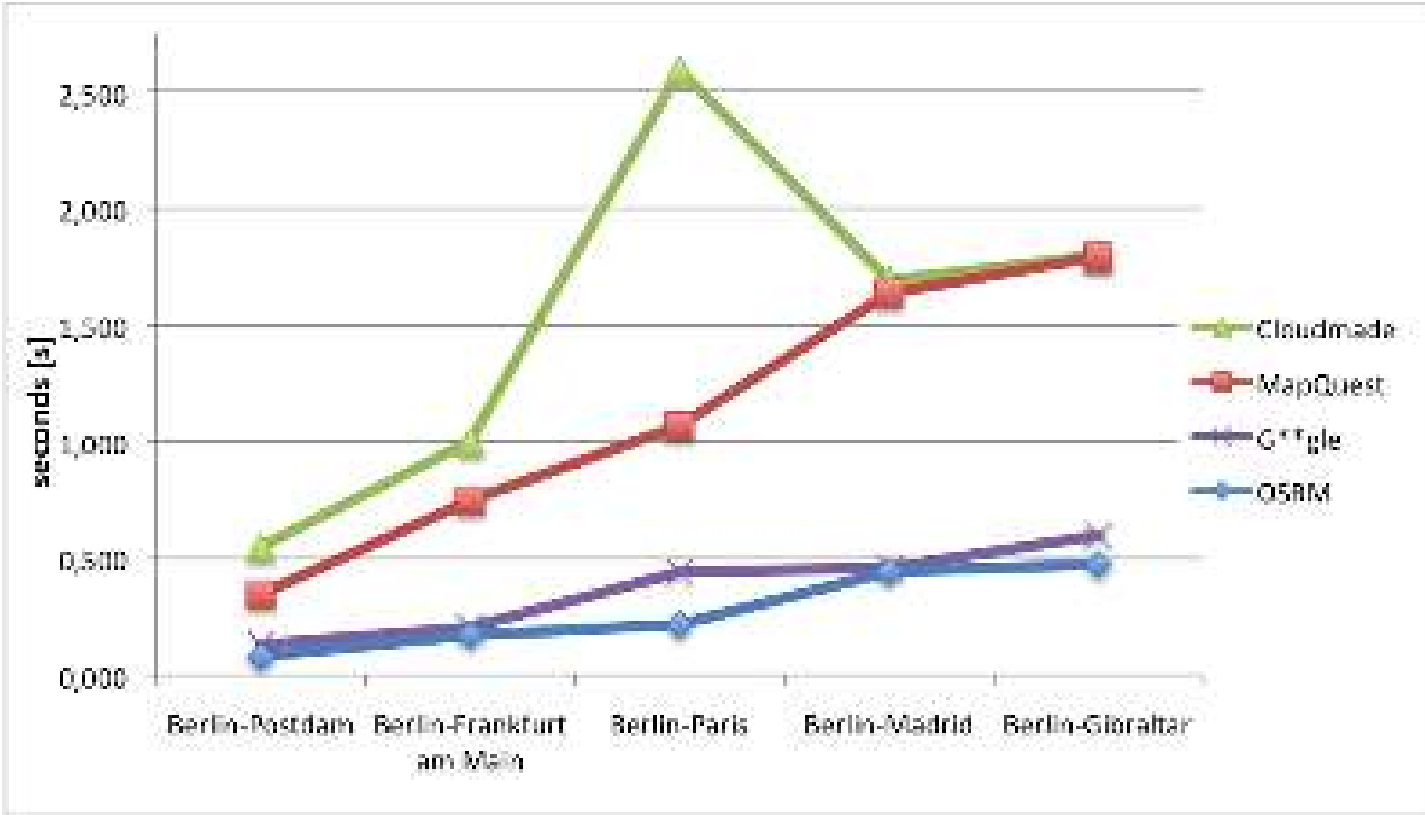
Found this on a blog:



General testimony: Project is great, but ...

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General testimony: Project is great, but ...



What's coming next

- Less RAM
- Even faster speed
- Alternative routes
- Self-hosting Javascript
- Print support
- Route sketches
- Distributed preprocessing
- Turn penalties
- Turn restrictions
- Distance matrices
- Multiple metrics

- Reachabilty Analyses
- Round-Trips
- General API
- Support for mobile devices
- Real-Time updates





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But: This is still a spare time project, so things will take time.

Thank you for the attention!



Questions?

- MoNav: `http://code.google.com/p/monav`
  0.3 Packages & Downloads: `http://monav.openstreetmap.de`
- OSRM: `http://project-osrm.org`
  Online Demo: `http://map.project-osrm.org`